



Email Header Assignment (Core)

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Coding Dojo Cybersecurity Boot Camp
11th February 2024



Email Headers (Core)



Learning Objectives:

- Analyze email headers.

Analyzing Email Headers

1. Download [this email header content](#).
2. Use your knowledge about email headers and [this lookup tool](#) to answer the following questions.
 - Who was the original sender of the email? (email address)
 - How many times was the email redirected before arriving at the destination?
 - What is the IP address from where the message originated?
 - What country did the email originate from?
 - Does it match the country indicated in the email message body?
 - What is the reply-to address? Is it different from the sender?

Assignment Submission

Upload a single document containing the answers to all of the questions above. Include any screenshots of any tools, or Google searches that you used in order to answer the questions when applicable.

Assignment Solution:

1. [this email header content](#) downloaded.

Email Header

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Received: from BL2PRD0711HT001.namprd07.prod.outlook.com (10.255.104.164) by
BY2PRD0711HT003.namprd07.prod.outlook.com (10.255.88.166) with Microsoft SMTP
Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:35 +0000

Received: from BL2PRD0711HT002.namprd07.prod.outlook.com (10.255.104.165) by
BL2PRD0711HT001.namprd07.prod.outlook.com (10.255.104.164) with Microsoft
SMTP Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:34 +0000

Received: from mail240-tx2-R.bigfish.com (65.55.88.116) by
BL2PRD0711HT002.namprd07.prod.outlook.com (10.255.104.165) with Microsoft
SMTP Server (TLS) id 14.16.257.4; Thu, 17 Jan 2013 23:35:34 +0000

Received: from mail240-tx2 (localhost [127.0.0.1]) by mail240-tx2-R.bigfish.com (Postfix)
with ESMTP id A05C032025F for <jerryp@mail.unomaha.edu>; Thu, 17 Jan 2013 23:35:33
+0000 (UTC)

X-Forefront-Antispam-Report:

CIP:59.125.100.113;KIP:(null);UIP:(null);IPV:NLI;H:bf.shako.com.tw;RD:59-125-100-113.HINET-
IP.hinet.net;EFVD:NLI

X-BigFish:

ps73(zz7f52hd926hzzlee6h1de0hlce5h1202hle76h1dlah1d2ahz58hz8275bhz2ei2a8h668h83
9h940h10d2h1177h1288h12a5h12a9h12bdh137ah139eh13b6h13eah144h1537h162dh163h1758h
17f1h184fh1898h300k503k953iwa7jk)

X-FOSE-spam: This message appears to be spam.

X-SpamScore: 73

Received-SPF: neutral (mail240-tx2: 59.125.100.113 is neither permitted nor denied by
domain of aol.com) client-ip=59.125.100.113; envelope-from=vieria@aol.com;
helo=bf.shako.com.tw ;shako.com.tw ;

Received: from mail240-tx2 (localhost.localdomain [127.0.0.1]) by mail240-tx2
(MessageSwitch) id 1358465731454940_30539; Thu, 17 Jan 2013 23:35:31 +0000
(UTC)

Received: from TX2EHSMHS007.bigfish.com (unknown [10.9.14.242]) by mail240-
tx2.bigfish.com (Postfix) with ESMTP id 675424200E7 for <jerryp@mail.unomaha.edu>; Thu,
17 Jan 2013 23:35:31 +0000 (UTC)

Received: from bf.shako.com.tw (59.125.100.113) by TX2EHSMHS007.bigfish.com
(10.9.99.107) with Microsoft SMTP Server (TLS) id 14.1.225.23; Thu, 17 Jan
2013 23:35:28 +0000

Received: from mail.shako.com.tw (59-125-100-112.HINET-IP.hinet.net
[59.125.100.112]) by bf.shako.com.tw (8.14.3/8.14.3) with ESMTP id
r0HNYCgA013928; Fri, 18 Jan 2013 07:34:12 +0800

X-Authentication-Warning: bf.shako.com.tw: Host 59-125-100-112.HINET-IP.hinet.net [59.125.100.112] claimed to be mail.shako.com.tw
 Authenticated-By: nobody
 X-SpamFilter-By: BOX Solutions SpamTrap 3.5 with qID r0HNXZSI028539, This message is passed by code: ctdos35128
 Received: from User (85-250-54-29.bb.netvision.net.il[85.250.54.29]) (authenticated bits=0)
 by mail.shako.com.tw (8.14.3/8.14.3/4.90) with ESMTP id r0HNXZSI028539; Fri, 18 Jan 2013 07:33:38 +0800
 X-BOX-Message-Id: r0HNXZSI028539
 Message-ID: <201301172333.r0HNXZSI028539@mail.shako.com.tw>
 X-Authentication-Warning: mail.shako.com.tw: Host 85-250-54-29.bb.netvision.net.il[85.250.54.29] claimed to be User
 Reply-To: <carr444@yahoo.com>
 From: JOSEPH CAMARAH VIEIRA <vieria@aol.com>
 Subject: [Spam-Mail] Dear Sir/Madam. (This message should be blocked: ctdos35128)
 Date: Fri, 18 Jan 2013 01:46:07 +0200
 Content-Type: text/plain; charset="Windows-1251"
 Content-Transfer-Encoding: 7bit
 X-Mailer: Microsoft Outlook Express 6.00.2600.0000
 X-MimeOLE: Produced By Microsoft MimeOLE V6.00.2600.0000
 To: Undisclosed recipients;;
 Return-Path: vieria@aol.com
 X-MS-Exchange-Organization-SCL: 7
 X-MS-Exchange-Organization-AVStamp-Mailbox: MSFTFF;1;0;0 0 0
 X-MS-Exchange-Organization-AuthSource: BL2PRD0711HT002.namprd07.prod.outlook.com
 X-MS-Exchange-Organization-AuthAs: Anonymous
 MIME-Version: 1.0

Dear Sir/Madam,
 my name is Joseph Camarah Vieira, i am from Guinea Bissau, my late father was the former minister of mines in my country Guinea Bissau, he was short dead by the rebels in my country, before his death he deposited \$60 million Dollars with Global Trust Security Company Accra Ghana, i want you to help me receive this money in your country for investment in your country i will give you 30% of the total sum when the funds arrive your country.

Regards.
 Mr Joseph Camarah Vieira
 00233 244 617 863
 my email:carr444@yahoo.com

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2. Use your knowledge about email headers and [this lookup tool](#) to answer the following questions.

a. **Who was the original sender of the email? (email address)** –

The email address of the original sender of the email is:

viera@aol.com. Identified from the excerpt below:

Received-SPF: neutral (mail240-tx2: 59.125.100.113 is neither permitted nor denied by domain of aol.com) client-ip=59.125.100.113; envelope-from=viera@aol.com;

b. **How many times was the email redirected before arriving at the destination?** – *From the email header provided, the mail was redirected 9 times. Excerpt below:*

- I. **Received:** from BL2PRD0711HT001.namprd07.prod.outlook.com (10.255.104.164) by BY2PRD0711HT003.namprd07.prod.outlook.com (10.255.88.166)
- II. **Received:** from BL2PRD0711HT002.namprd07.prod.outlook.com (10.255.104.165) by BL2PRD0711HT001.namprd07.prod.outlook.com (10.255.104.164)
- III. **Received:** from mail240-tx2-R.bigfish.com (65.55.88.116) by BL2PRD0711HT002.namprd07.prod.outlook.com (10.255.104.165)
- IV. **Received:** from mail240-tx2 (localhost [127.0.0.1]) by mail240-tx2-R.bigfish.com (Postfix)
- V. **Received:** from mail240-tx2 (localhost.localdomain [127.0.0.1]) by mail240-tx2 (MessageSwitch)
- VI. **Received:** from TX2EHSMHS007.bigfish.com (unknown [10.9.14.242]) by mail240-tx2.bigfish.com (Postfix)
- VII. **Received:** from bf.shako.com.tw (59.125.100.113) by TX2EHSMHS007.bigfish.com (10.9.99.107)
- VIII. **Received:** from mail.shako.com.tw (59-125-100-112.HINET-IP.hinet.net [59.125.100.112]) by bf.shako.com.tw
- IX. **Received:** from mail.shako.com.tw (85-250-54-29.bb.netvision.net.il [85.250.54.29]) by bf.shako.com.tw

c. What is the IP address from where the message originated?

The IP address from where the message originated is "**59.125.100.113**". This IP address is identified in the "Received-SPF" line (of the email header), that contains the information about the sender's system. Excerpt below:

*Received-SPF: neutral (mail240-tx2: **59.125.100.113** is neither permitted nor denied by domain of aol.com) client-ip=59.125.100.113; envelope-from=**vieria@aol.com**; helo=bf.shako.com.tw ;shako.com.tw*

d. What country did the email originate from?

Using a tool to lookup the sender's IP address **59.125.100.113**, it is likely that the email originated from **Taiwan**.

Geolocation data from IP2Location (Product: DB6, 2024-2-1)

 IP ADDRESS: 59.125.100.113	 ISP: Chunghwa Telecom Co. Ltd.
 COUNTRY: Taiwan 	 ORGANIZATION: Not available
 REGION: Taipei	 LATITUDE: 25.0382
 CITY: Taipei	 LONGITUDE: 121.5636

Geolocation data from ipinfo.io (Product: API, real-time)

 IP ADDRESS: 59.125.100.113	 ISP: Not available
 COUNTRY: Taiwan 	 ORGANIZATION: AS3462 Data Communication Business Group
 REGION: Taiwan	 LATITUDE: 25.0478
 CITY: Taipei	 LONGITUDE: 121.5319

Geolocation data from DB-IP (Product: API, real-time)

 IP ADDRESS: 59.125.100.113	 ISP: Chunghwa Telecom Co., Ltd.
 COUNTRY: Taiwan 	 ORGANIZATION: Chunghwa Telecom Co. Ltd.
 REGION: Taiwan	 LATITUDE: 25.033
 CITY: Taipei	 LONGITUDE: 121.565

e. Does it match the country indicated in the email message body?

No, it doesn't.

The likely country of email origin is **Taiwan**, which doesn't match '**Guinea Bissau**' indicated in the body of the email.

f. What is the reply-to address? Is it different from the sender?

The "Reply-To" address provided in the email header is "**carr444@yahoo.com**," which is different from the sender's address ("**vieria@aol.com**").

This means that any reply to the email will be directed to the Yahoo email address instead of the AOL address.